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[22CE52]

FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER-2024

GEOTECHNICAL ENGINEERING

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A	**5 x2=10**		
			BL	**CO**	**Marks**
1.	**a.**	Define void ratio and explain its significance in soil mechanics.	2	1	**02**
	b.	What is consolidation? Differentiate between primary and secondary consolidation.	2	2	**02**
	c.	State the principle of effective stress in soil mechanics.	2	3	**02**
	d.	List different types of soil samplers used in field soil investigation.	2	4	**02**
	e.	Explain punching shear failure and its causes.	2	5	**02**
		PART-B	**6 X 5=30**		
2.	**a.**	Explain the laboratory procedure for the determination of Atterberg limits.	2	1	**05**
		OR			
	b.	Write short notes on the significance of specific gravity in soil classification.	2	1	**05**
3.	**a.**	Differentiate between Standard Proctor and Modified Proctor tests.	2	2	**05**
		OR			
	b.	Explain the procedure for the core cutter method of determining field density.	2	2	**05**
4.	**a.**	Explain the process of determining the shear strength of soil using a direct shear test.	2	3	**05**
		OR			
	b.	Differentiate between Unconfined Compression and Triaxial Compression Tests.	3	3	**05**
5.	**a.**	Discuss the various types of foundations used for different soil conditions.	2	4	**05**
		OR			
	b.	Explain the planning of soil exploration programs.	2	4	**05**
6.	**a.**	Explain Terzaghi's bearing capacity theory and its limitations.	2	5	**05**

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		OR			
	b.	Differentiate between the plate load test and the Standard Penetration Test (SPT).	3	5	**05**
7.	**a.**	Define settlement. Discuss the types and causes of settlement.	2	4	**05**
		OR			
	b.	Explain the factors that affect settlement in shallow foundations.	2	5	**05**
		PART-C	**6 X 10=60**		
8.	**a.**	Given a soil sample with a void ratio of 0.5 and a total volume of 1 m^3, calculate the volume of voids and the volume of solids. How does the void ratio influence the soil's permeability and drainage characteristics	3	1	**10**
		OR			
	b.	A foundation is subjected to a vertical load of 100 kN. Calculate the settlement of the foundation if the compressibility coefficient is 1.5×10^{-6} m^2/kN. Discuss the influence of soil type on settlement characteristics.	4	4	**10**
9.	**a.**	Explain how the coefficient of permeability can be determined in the laboratory. Discuss its significance in the design of drainage systems.	2	2	**10**
		OR			
	b.	Calculate the bearing capacity of a strip footing of width 2 m, located at a depth of 1 m in a soil layer with a cohesion of 25 kPa and an angle of internal friction of 30 degrees. Explain the assumptions made in the bearing capacity calculation.	3	5	**10**
10.	**a.**	Analyze a scenario where a building experiences differential settlement. What factors might contribute to this condition, and how can they be mitigated?	4	4	**10**
		OR			
	b.	Calculate the maximum load a soil can support given a bearing capacity of 150 kPa and an area of footing of 3 m^2. Assess the potential risks associated with exceeding this load.	4	5	**10**
11.	**a.**	Discuss the importance of field penetration tests in soil investigation and describe how the Standard Penetration Test (SPT) is conducted, including its limitations.	2	2	**10**

		OR			
	b.	Given a soil sample with a moisture content of 20%, determine its dry density if the saturated density is 20 kN/m³. Discuss how moisture content impacts the engineering properties of the soil.	4	1	**10**
12.	**a.**	Explain the concept of seepage flow and its relevance in geotechnical engineering, particularly in the context of slope stability and erosion.	3	2	**10**
		OR			
	b.	Describe the spring analogy used for understanding primary consolidation, including its practical applications in predicting settlement behavior.	3	2	**10**
13.	**a.**	A soil sample is tested using a direct shear test, yielding a maximum shear stress of 50 kPa and a normal stress of 100 kPa. Calculate the friction angle using the Mohr-Coulomb theory. Discuss how this angle influences slope stability analyses.	4	3	**10**
		OR			
	b.	Discuss the implications of water table fluctuation on the bearing capacity of soil, providing examples of real-world cases where fluctuations have caused structural failures.	3	4	**10**